Build Your Own SDR Doppler Radar

A student guide to the hardware, the GNU Radio and Python software and the workflow

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What you will learn

- How a continuous-wave (CW) Doppler radar measures speed, and how to compute it.
- How to build an SDR flowgraph in GNU Radio and program the PlutoSDR from Python.
- Digital signal processing: mixing, decimation, filtering, windowing, FFT and peak detection.
- How to build an Arduino display and buzzer circuit and talk to it over USB serial.
- How to test, validate and present an engineering project.



The ADALM-PLUTO with its Tx and Rx antennas.

1. Theory in five minutes

The radar transmits a constant tone. A target moving with radial speed v reflects it with a small frequency shift fd = 2·v·fc/c. The receiver uses the same local oscillator as the transmitter, so after down-conversion only fd is left: a low audio-range tone whose frequency is proportional to speed.

Worked example: a car at 50 km/h = 13.89 m/s. fd = 2 × 13.89 × 2.482×10^{9} / 3×10^{8} = **229.8 Hz**. Measuring 229.8 Hz in the spectrum therefore means 50 km/h.

Concept	Why it matters in this project
IQ (complex) samples	The sign of the frequency tells the direction: + approaching, - receding
Tone offset (+100 kHz)	Keeps the echo away from the receiver's DC offset and LO leakage at 0 Hz
Resolution = fs / N	8000 Hz / 2048 = 3.9 Hz per bin = 0.85 km/h (0.1 km/h after interpolation)
Observation time	2048 samples at 8 kS/s = 0.26 s; longer = finer bins but slower response
Cosine effect	Only the speed along the line of sight is measured; aim along the target's path
Clutter	Fans and vibration give mirror peaks at +fd and -fd; real targets give one side only

2. Parts list

Qty	Part	Notes
1	ADALM-PLUTO SDR	Firmware v0.39; use the middle (data) micro-USB port
2	2.4 GHz antennas, SMA male	Router antennas are often RP-SMA (no centre pin) and need an adapter
1	SMA extension cable, male-female, 0.5-1 m	Separates Tx and Rx antennas
1	Arduino Nano + USB cable	Clone boards use the CH340 USB chip
1	TM1637 4-digit LED module	Pins CLK, DIO, VCC (5V), GND
1	Passive buzzer (5 V)	Must be passive: an active buzzer cannot change pitch
1	NPN transistor 2N2222 or BC547	Switches the buzzer current
1	1 kΩ resistor	Base resistor
1	1N4148 or 1N4007 diode	Protects the transistor (magnetic buzzers)
1	Half-size breadboard + jumper wires	

3. Hardware build

3.1 Radar front end

- Screw one antenna onto **Tx**. Connect the extension cable to **Rx** and the second antenna to its end.
- Stand both antennas upright and about 0.5 m apart, facing the area where targets move.
- Check the connector: an SMA male antenna has a **centre pin**. A hole means RP-SMA, which screws on but makes no contact.
- Plug the Pluto into the laptop with the middle micro-USB port and wait about 30 s for the Ready LED.

3.2 Arduino, display and buzzer

Place the Nano across the centre gap of the breadboard, one pin row on each side. Connect Nano 5V to the + rail and GND to the - rail, then wire:

From	To
TM1637 CLK	Nano D2
TM1637 DIO	Nano D3
TM1637 5V / GND	+ rail / - rail
Nano D9	1 kΩ resistor → NPN base
NPN emitter	- rail (GND)
NPN collector	buzzer (-) and diode anode
buzzer (+)	+ rail (5V) and diode cathode (stripe)

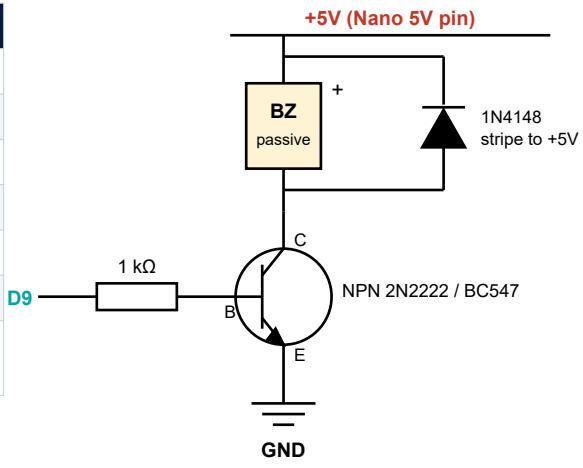


Figure 1 - Buzzer driver: D9 switches the NPN transistor; the diode absorbs the buzzer's back-EMF.

Transistor pins (flat face toward you, legs down): **2N2222 = E B C**, **BC547 = C B E**. The two types are reversed. **Before powering:** check that no wire joins + and - directly, and that 5V and GND of the display are not swapped.

4. Software setup

Step	What to install	Check
1	Python 3.10+ from python.org (tick "Add Python to PATH")	python --version
2	PlutoSDR USB driver and libiio v0.26 (in drivers/)	Pluto appears as a USB drive
3	Python packages: pip install -r requirements.txt	pyadi-iio, pylibiio, numpy, scipy, matplotlib, pyserial, sounddevice
4	Arduino IDE 2 + library "TM1637" by Avishay Orpaz	Board: Arduino Nano, Old Bootloader for clones
5	radioconda (GNU Radio 3.10 + PlutoSDR blocks + its own Python); untick Add to PATH	Start menu: GNU Radio Companion

Project files

File	Purpose
01_pluto_test.py	Hardware check: transmits a tone and shows the received spectrum
02_doppler_radar.py	Python version of the radar (backup): window, laptop alarm; uses doppler_core.py
03_alarm_test.py	Sends simulated passes to the Arduino (no Pluto needed)
arduino_port.py	Finds the Arduino's COM port automatically
speed_alarm/speed_alarm.ino	Arduino sketch: serial parser, LED display, buzzer PWM
gnuradio/doppler_radar.grc	GNU Radio flowgraph: Pluto TX/RX, filters, displays, sliders
gnuradio/doppler_camera_block.py	Source of the Doppler Speed Camera block (embedded Python)
doppler_core.py	Detector, speed, pass tracking and logging shared by GNU Radio and Python
START_GNURADIO.bat / EDIT_GNURADIO.bat	Run the GNU Radio radar / open it in GNU Radio Companion
START_RADAR.bat / TOOLS.bat	Python radar / optional menu with tests, upload and drivers

5. GNU Radio flowgraph

Open **gnuradio/doppler_radar.grc** in GNU Radio Companion (EDIT_GNURADIO.bat). Each block does one step of the radar; the sliders are QT GUI Range variables, so moving one immediately re-tunes the Pluto or changes the detector. The radar-specific logic is one Embedded Python Block, which calls doppler_core.py - the same code the Python version uses.

GNU Radio block	Setting	Purpose
Signal Source	100 kHz complex tone, amplitude 0.5	CW waveform (offset from 0 Hz)
PlutoSDR Sink	LO = Frequency slider, 1 MS/s, TX attenuation slider	Transmits the CW tone
PlutoSDR Source	LO = Frequency slider, 1 MS/s, manual RX gain slider	Receives TX leakage + echoes
Frequency Xlating FIR Filter	shift -100 kHz, low-pass, decimate 25	Tone to 0 Hz, 1 MS/s to 40 kS/s
FIR Filter	low-pass 3.5 kHz, decimate 5	40 kS/s to 8 kS/s Doppler signal
Doppler Speed Camera (embedded Python)	Max speed, Speed limit and Min signal sliders	FFT, clutter map, detection, km/h, capture, log, Arduino
QT GUI Frequency Sink / Waterfall	8 kHz span, 4096 / 1024 points	Live Doppler spectrum
Function Probe + QT GUI Label	polled 5 times per second	LIVE SPEED and LAST SPEED readouts
QT GUI Range (6 sliders)	Frequency, Max speed, Speed limit, RX gain, TX attenuation, Min signal	Live adjustment

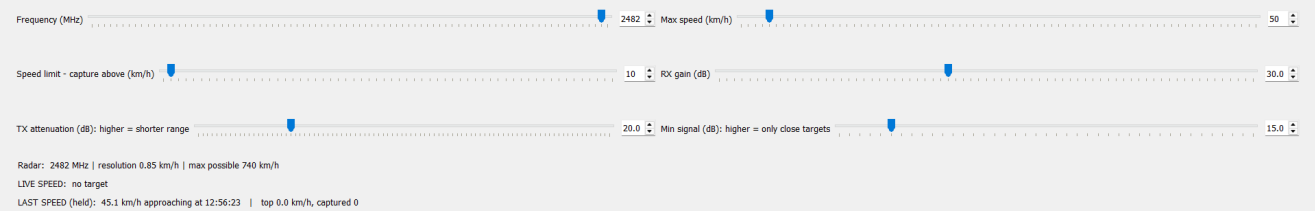


Figure 2 - GNU Radio radar window: sliders and speed readouts.

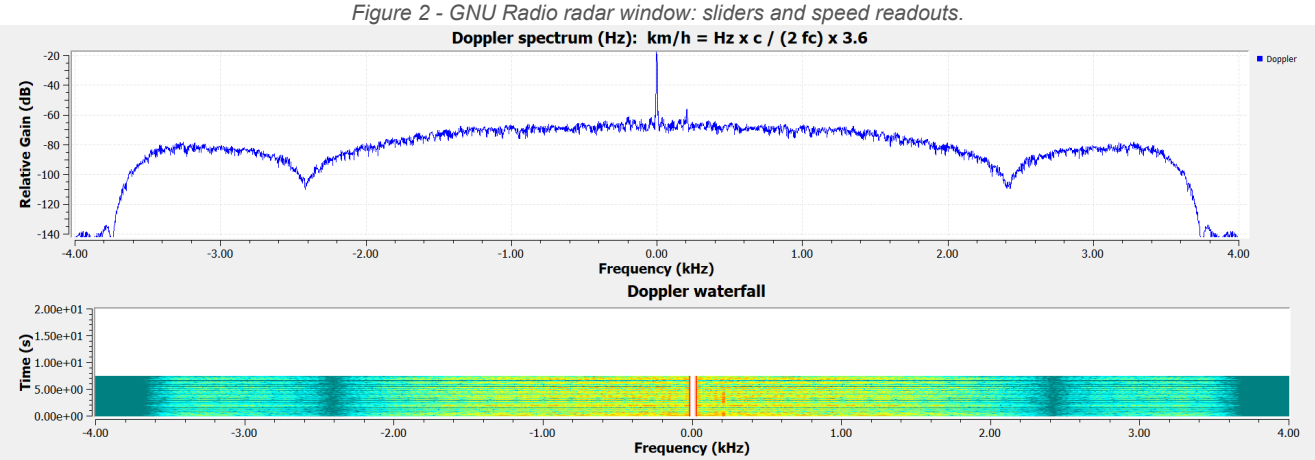


Figure 3 - Doppler spectrum and waterfall with a 45 km/h test target.

Default settings

Slider (block ID)	Default	Meaning
Frequency (freq_mhz)	2482 MHz	Carrier; 2400-2483 MHz with the 2.4 GHz antennas
Max speed (max_kmh)	50 km/h	Highest speed measured; up to 740 km/h at 2482 MHz
Speed limit (limit_kmh)	10 km/h	Faster passes are CAPTURED: logged and kept blinking on the LED
TX attenuation (tx_atten)	20 dB	Higher = less power = shorter range (x1.8 range per 10 dB)
Min signal (min_signal)	15 dB	Signal a target needs above the noise; higher = only closer targets
RX gain (rx_gain)	30 dB	As high as possible without the CLIPPING warning

Sliders change the running radar at once but are not saved. To change a default: open EDIT_GNURADIO.bat, double-click the slider block (its ID is in the table), set Default Value, press OK and Ctrl+S, then start the radar again.

gnuradio/doppler_camera_block.py (shortened)

```
class blk(gr.sync_block):
    def __init__(self, freq_mhz=2482.0, max_kmh=50.0, limit_kmh=10.0, ..., min_signal_db=15.0):
        gr.sync_block.__init__(self, name="Doppler Speed Camera", in_sig=[np.complex64], out_sig=None)

    def work(self, input_items, output_items):
        self._apply_settings()                      # slider moved? re-tune detector / re-learn
        self.buf = np.concatenate([self.buf, input_items[0]])
        while len(self.buf) >= dc.FRAME:            # 512 samples = 64 ms at 8 kS/s
            frame, self.buf = self.buf[:dc.FRAME], self.buf[dc.FRAME:]
            p_db, thr_db, noise_db, k, snr, confirmed = self.det(frame)
            kmh, proximity, ended = self.cam.update(self.det, p_db, k, snr, confirmed)
        return len(input_items[0])
```

Max speed is a search limit: the 8 kS/s chain passes Doppler up to about 3.4 kHz, i.e. 740 km/h at 2482 MHz (v = fd·c/2fc). A lower carrier gives a smaller Doppler shift, so the screen shows the resolution and the highest possible speed for the current frequency.

6. Code walkthrough

In the GNU Radio version the PlutoSDR Sink/Source and the two filter blocks do steps 6.1 and 6.2; the Python version does the same in code, shown here because it makes each step visible. Steps 6.3 to 6.5 are doppler_core.py, used by both versions.

6.1 Configure the PlutoSDR and start transmitting

The pyadi-iio library exposes the Pluto as an object. Both oscillators are set to the same frequency, the gains are fixed, and a cyclic buffer makes the Pluto repeat the tone forever.

02_doppler_radar.py (Python version) - start_pluto() and main()

```
sdr = adi.Pluto("ip:192.168.2.1")
sdr.sample_rate = 1_000_000
sdr.rx_lo = sdr.tx_lo = int(2.482e9)          # same LO for TX and RX
sdr.gain_control_mode_chan0 = "manual"
sdr.rx_hardwaregain_chan0 = 30                # dB, lower if CLIPPING
sdr.tx_hardwaregain_chan0 = -20               # dB attenuation (TX attenuation slider)
sdr.rx_buffer_size = 64000                    # 64 ms per read

n = np.arange(64000)                          # 6400 whole periods of 100 kHz
sdr.tx_cyclic_buffer = True
sdr.tx(0.5 * np.exp(2j * np.pi * 100e3 * n / 1e6) * 2**14)
```

Tip: the buffer holds a whole number of tone periods, so the repeated tone has no phase jump.

6.2 Down-conversion and decimation

Multiplying by a complex exponential shifts the 100 kHz tone to 0 Hz. Averaging blocks of 25 samples is a cheap low-pass filter whose nulls fall on multiples of 40 kHz; an FIR filter then removes everything above 3.5 kHz before keeping every 5th sample.

02_doppler_radar.py (Python version) - Downconverter; GNU Radio: Frequency Xlating FIR + FIR Filter

```
class Downconverter:
    def __init__(self):
        self.mixer = np.exp(-2j * np.pi * TONE_HZ * np.arange(RX_BLOCK) / SAMPLE_RATE)
        self.h2 = dsp.firwin(101, 3.5e3, fs=SAMPLE_RATE / DECIM_1)
        self.z2 = np.zeros(len(self.h2) - 1, complex)      # filter state between blocks

    def __call__(self, x):
        x = x * self.mixer                                 # tone -> 0 Hz
        y = x.reshape(-1, DECIM_1).mean(axis=1)            # 1 MS/s -> 40 kS/s
        y, self.z2 = dsp.lfilter(self.h2, 1, y, zi=self.z2)
        return y[::DECIM_2]                                # 40 kS/s -> 8 kS/s
```

6.3 Spectrum and detection

The newest 512 samples are appended to a 2048-sample history (0.26 s). The mean is removed (direct TX-to-RX leakage is a constant), a Hann window reduces spectral leakage, and the FFT gives the spectrum. Each frequency bin gets its own threshold: at least Min signal above the noise floor, and above a **clutter map** - the mean plus 3 dB plus two standard deviations that the bin showed while nothing moved. The map ignores the Pluto's fixed spurs and steady background lines. Of the five strongest peaks, the first that also beats its mirror frequency (fans give both +fd and -fd) is the target; it must be found in two frames in a row.

doppler_core.py - Detector (shortened)

```
self.history = np.concatenate([self.history[len(y):], y])
x = self.history - self.history.mean()             # remove TX leakage (DC)
spec = np.fft.fftshift(np.fft.fft(x * self.window)) / NFFT
p_db = 10 * np.log10(np.abs(spec) ** 2 + 1e-30)

noise_db = np.median(p_db[band])                   # noise floor
thr_db = np.maximum(noise_db + self.detect_db,     # Min signal slider
                    self.clutter_db + CLUTTER_DB + CLUTTER_K * np.sqrt(self.clutter_var))
excess = p_db - thr_db
for c in idx[np.argsort(excess[idx])[::-1][:CANDIDATES]]:   # 5 strongest peaks
    if excess[c] < 0:
        break
    km = NFFT - c                                  # mirror bin (-fd)
    if p_db[c] - p_db[max(km - 2, 0):km + 3].max() >= MIRROR_DB:
        k, hit = c, True
        break
self.hits = self.hits + 1 if hit else 0            # confirm over 2 frames
```

6.4 Sub-bin precision

A parabola through the peak bin and its two neighbours finds the true peak between bins. On a simulated target this reduced the error from up to 0.2 km/h to below 0.01 km/h.

doppler_core.py

```
def peak_offset(p_db, k):
    a, b, c = p_db[k - 1], p_db[k], p_db[k + 1]
    curve = a - 2 * b + c
    return float(np.clip(0.5 * (a - c) / curve, -0.5, 0.5)) if curve < 0 else 0.0

fd = freqs[k] + peak_offset(p_db, k) * FS_DOPPLER / NFFT
kmh = fd * C / (2 * fc) * 3.6                    # doppler_to_kmh(fd, fc)
```

6.5 Pass tracking and capture

While a target is present the program keeps the highest speed of the pass. After about 0.4 s without a target the pass ends: its peak speed is held on the display, written to speed_log.csv, and captured if it is above the speed limit. Together with the 0.26 s window, the held speed reaches the LED about 0.75 s after the target has left (measured on the Pluto).

doppler_core.py - SpeedCamera.update

```
if proximity > 0:                                # target present in this frame
    if abs(kmh) > abs(self.pass_peak):
        self.pass_peak = kmh                     # keep the peak speed of the pass
    self.pass_frames += 1
    self.lost_frames = 0
elif self.pass_frames:
    self.lost_frames += 1
    if self.lost_frames >= PASS_END_FRAMES:      # 6 frames = ~0.4 s without target
        ended = self._end_pass()                 # hold, log, capture if > limit
```

6.6 Laptop → Arduino protocol

Every frame (64 ms) one text line is sent at 115200 baud: `S45.3,P70,V75,L38.0,F0` = live speed, proximity %, volume %, last speed (-1 = none) and the capture flag. If no line arrives for 0.5 s the Arduino goes silent.

6.7 Arduino sketch

The buzzer uses Timer1 in fast-PWM mode on pin D9: the timer's TOP value sets the pitch and the compare value sets the duty cycle (loudness). The beep pattern comes from millis().

speed_alarm.ino

```
void buzzerOn(float pitchHz, int vol) {
  uint16_t top = (uint16_t)(2000000.0 / pitchHz) - 1;   // 16 MHz / prescaler 8
  float duty = 0.02 + 0.48 * vol / 100.0;               // 2 % .. 50 %
  ICR1 = top;
  OCR1A = (uint16_t)(top * duty);
  TCCR1A = _BV(COM1A1) | _BV(WGM11);                    // fast PWM, TOP = ICR1
  TCCR1B = _BV(WGM13) | _BV(WGM12) | _BV(CS11);
}

float beepsPerSec = 1.0 + 9.0 * proximity / 100.0;      // closer = faster beeps
unsigned long period = (unsigned long)(1000.0 / beepsPerSec);
bool beepOn = (millis() % period) < period / 2;
float pitch = 600.0 + 6.0 * volume;                     // 600 .. 1200 Hz
```

6.8 Safe shutdown

A radar must never keep transmitting after the program ends. All work runs inside `try ... finally`; on any exit (window closed, Ctrl+C, error) the TX buffer is destroyed and the TX attenuation is set to -89 dB.

7. Build and test workflow

Build in stages and test each stage before adding the next. When something fails you then know which stage caused it.

Stage	Action	Pass criterion
1	Install Python, drivers and packages	python -c "import adi" runs without error
2	Connect Pluto and antennas; run 01_pluto_test.py	Strong peak at +100 kHz, below 0 dBFS
3	Run the radar on the laptop only; wave a hand	Speed shown, direction sign correct
4	Wire the display; upload speed_alarm.ino	Display shows 88:88, then ----
5	Run 03_alarm_test.py	Display shows 15, 35, then 72 blinking
6	Add the buzzer circuit	Start-up chirp; beeps during the alarm test
7	Full system: START_GNURADIO.bat (sliders, display)	Walking and vehicle passes shown, held and logged
8	Validation (section 8)	Error table filled in for the report

8. Testing and validation

- **Reference speed:** mark 10 m on the ground and time each pass with a stopwatch; reference = distance / time. For vehicles, use the speedometer at a constant speed.
- Walk or drive **straight toward the antennas** to avoid the cosine error.
- Repeat 5 times per speed (slow walk, fast walk, jog, vehicle) and compare with speed_log.csv.
- Run 5 minutes in an empty room to count false alarms. Fans and moving objects must be out of view.
- If CLIPPING appears, increase the antenna separation or lower --rx-gain.

Run	Distance (m)	Time (s)	Reference (km/h)	Radar (km/h)	Error (%)
1	10				
2	10				
3	10				

9. Troubleshooting

Symptom	Cause and fix
PlutoSDR not found / Open unlocked: -16	Wrong USB port, drivers missing, or another radar window still open
No peak in 01_pluto_test.py	RP-SMA antenna on an SMA port (no contact): use an adapter
CLIPPING on screen	Antennas too close or RX gain too high
Targets with nobody moving	Ceiling fan or vibrating objects in view
Upload error: not in sync	Clone Nano: choose ATmega328P (Old Bootloader)
Access is denied on the COM port	Arduino IDE or Serial Monitor still open
Display blank	CLK/DIO swapped or loose, or 5V/GND not connected
Buzzer silent	Transistor pins reversed (2N2222 vs BC547) or an active buzzer used

10. Presenting the project

Suggested slide order

- Problem and objective
- Doppler principle with the worked example
- System architecture (hardware block diagram)
- GNU Radio flowgraph and signal-processing chain
- Speed-camera logic, display and buzzer
- Results: validation table and screenshots
- Live demonstration
- Conclusion and future work

Demonstration script

- Show the hardware and explain the Tx/Rx antennas.
- Open EDIT_GNURADIO.bat to show the block diagram, then start START_GNURADIO.bat and explain the spectrum.
- Move the Frequency and Max speed sliders to show that the radar is software-defined.
- Walk toward the antennas: point out the live speed, the beeps and the held last speed.
- Do a fast pass over the limit: show the red CAPTURED message and the blinking LED.
- Open speed_log.csv to show the recorded passes.

Questions to prepare for

Question	Short answer
Why a tone offset of 100 kHz?	The receiver has a DC offset and LO leakage at 0 Hz; the offset keeps the echo away from it
What limits the resolution?	Observation time: 0.26 s gives 3.9 Hz bins; interpolation gives 0.1 km/h
What is the maximum speed?	The Max speed slider (default 50 km/h); the 8 kS/s chain allows ~740 km/h
Why is a toy car only seen close by?	A small target echoes little and the echo falls with distance^4
How are fans rejected?	A vibrating object gives peaks at +fd and -fd; the mirror test rejects it
Can it measure distance?	No, CW radar measures speed only; FMCW would add range
Why does an off-axis target read low?	Only the radial component v·cos(angle) shifts the frequency

11. Exercises

- Change NFFT to 1024 and 4096: how do resolution and response time change?
- Plot speed against time for a walking pass and identify arm and leg motion.
- Add a webcam snapshot when a pass is captured.
- Replace the dipoles with directional patch antennas and measure the range gain.